

Smart City Traffic Intelligence

Data Analytics Insights Report

Findings and implications for the Smart-City Mobility Analytics Team

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Overview

This report summarises the most important findings from the Part 1 data analytics work on westbound I-94 traffic near Minneapolis–St Paul. The analysis combined 40,575 deduplicated hourly traffic records (October 2012 to September 2018) with weather and US federal holiday data, using SQL, descriptive statistics, probability analysis, and an interactive Power BI dashboard. The purpose is to translate these findings into practical guidance for planning and operating the city's mobility network.

Traffic Volume Follows a Strong, Predictable Daily Cycle

Traffic volume averages 3,291 vehicles per hour but swings widely (standard deviation $\approx 1,985$, a coefficient of variation of about 60%), ranging from near-empty overnight to a peak of 7,280 vehicles per hour. This variability is not random: it is driven almost entirely by the daily commuting cycle, with a clear morning peak around 7am and a larger evening peak in the mid-afternoon to early evening, separated by a deep overnight trough. Year-over-year, average traffic per recorded hour was fairly stable from 2012 to 2015 before dipping to a six-year low in 2016 (3,194/hr) and then rising to a six-year high in 2017 (3,377/hr, +5.7%). Because data coverage varies sharply by year (from 23.9% of 2012 to 99.5% of 2017), this per-hour average, rather than raw yearly totals, is the reliable basis for year-on-year comparison.

Weather and Temperature Are Minor Factors, Not Major Drivers

Across every analytical lens used, weather showed only a weak relationship with traffic. Temperature correlates with traffic volume at just $r = 0.14$ (R^2 under 2%), and the scatter plot confirms no visible trend across the real temperature range of roughly -30°C to $+35^{\circ}\text{C}$. Congestion (hours above 5,500 vehicles) occurs 15.1% of the time overall, and while congestion and clear weather are not statistically independent, the effect is small: the odds of congestion in clear weather are about 26% lower than in cloudy weather. Average traffic does differ more noticeably by weather category on the dashboard, highest under Clouds, Haze, and Drizzle (roughly 3,000–3,620 vehicles/hour) and lowest under Squall (420 vehicles/hour), but Squall is a rare event backed by very few records, so that figure should be treated as directional rather than precise. Holiday-specific checks (New Year's Day and Labor Day, 2015–2017) reinforce the same conclusion: traffic and temperature did not move together consistently, in one case traffic rose 11.3% while temperature fell sharply, confirming that calendar and time-of-day effects outweigh weather.

Data Quality Issues Were Identified and Corrected

Several data quality issues were found during preparation and addressed before analysis: 5,445 hours had duplicate timestamp entries from repeated weather logging, which would have inflated yearly totals if left unmerged; the holiday column stored “None” as literal text rather than a true blank, masking how much data was actually missing; temperature, rainfall, and snowfall were auto-detected as text rather than numeric, which would have blocked correct charting; and a small cluster of records (about 0.2% of rows) logged temperature as 0 Kelvin, a sensor error rather than a real reading. All of these were cleaned or explicitly flagged, and where they could plausibly affect a result (such as the temperature correlation), the analysis was re-run without them to confirm the conclusion did not change.

Implications for the Smart-City Mobility Team

Because time-of-day and day-of-week patterns are the dominant driver of traffic, operational decisions, such as signal timing, transit frequency, and congestion-based measures, should be planned primarily around the commuting cycle rather than weather forecasts. Weather-responsive interventions are unlikely to meaningfully improve routine mobility outcomes and are better reserved for rare, severe events (like squalls), where a real but numerically fragile drop in traffic was observed. The Power BI dashboard built in this phase, with hour, year, weather, and traffic-category filters, gives the team a reusable, self-service view of these patterns rather than a one-off analysis. Finally, fixing data collection at the source, correcting the temperature sensor issue, deduplicating weather feed logs, and coding holidays consistently, would improve confidence in any future predictive modelling, and year-over-year comparisons should continue to account for uneven historical data coverage rather than relying on raw totals.

Next Steps

The cleaned dataset, engineered fields (Hour, Temperature in Celsius, Traffic Category), and the findings summarised here carry forward directly into Part 2 of this capstone, where a Python pipeline will formalise this cleaning and feature engineering, and Part 3, where these patterns inform predictive modelling for the smart-city mobility use case.